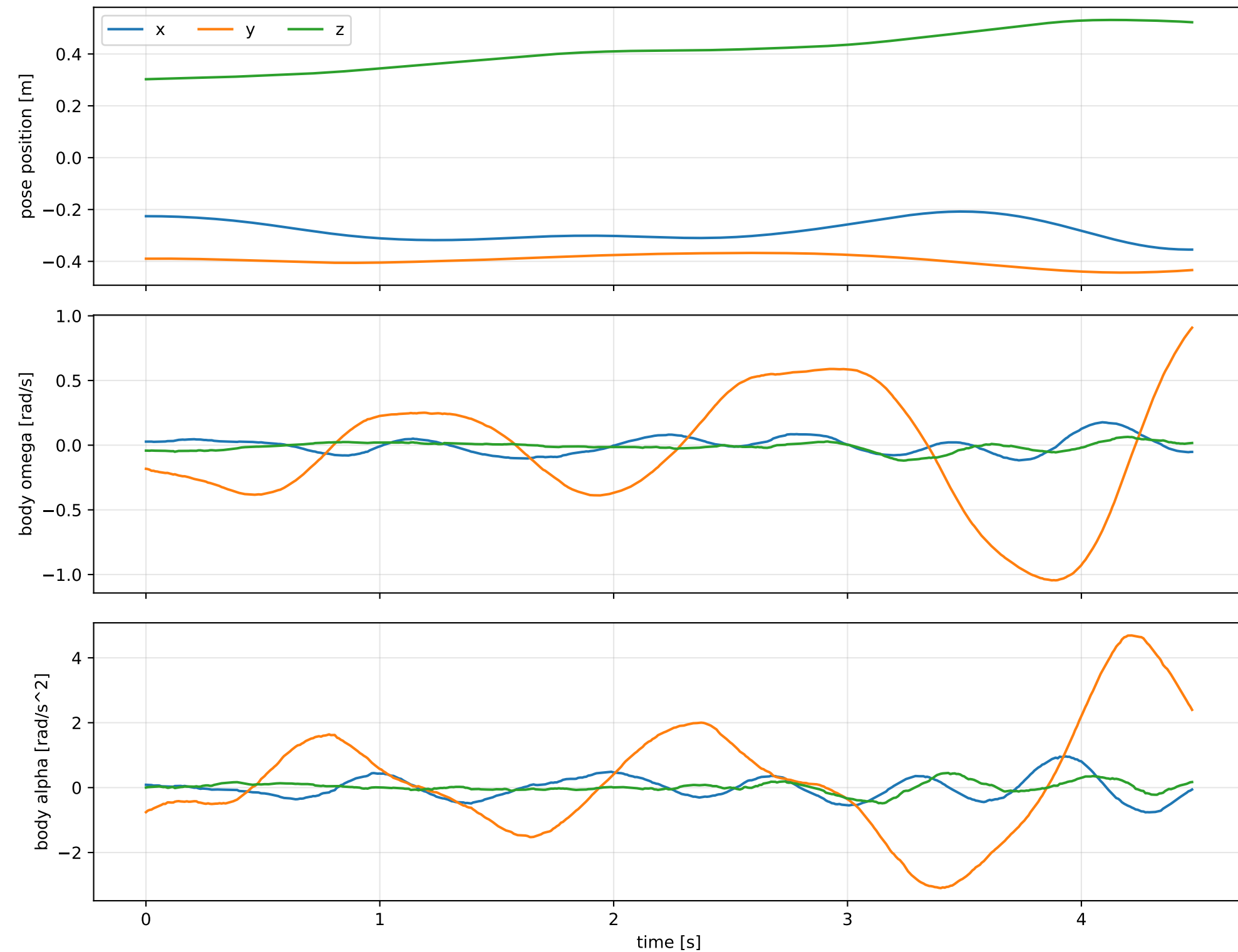
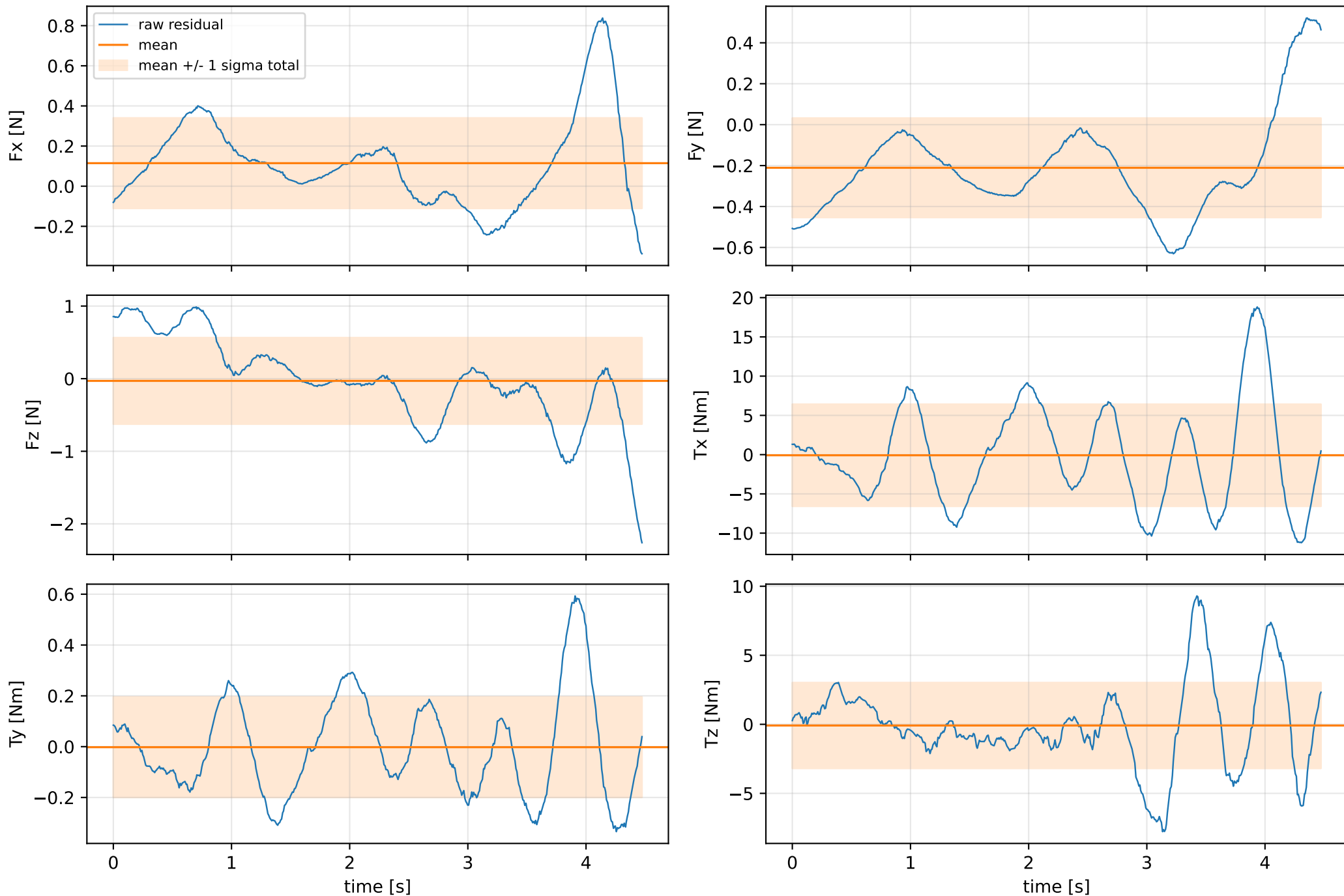


force_over_mass_rotor_2_nominal: SG trajectory and derived motion



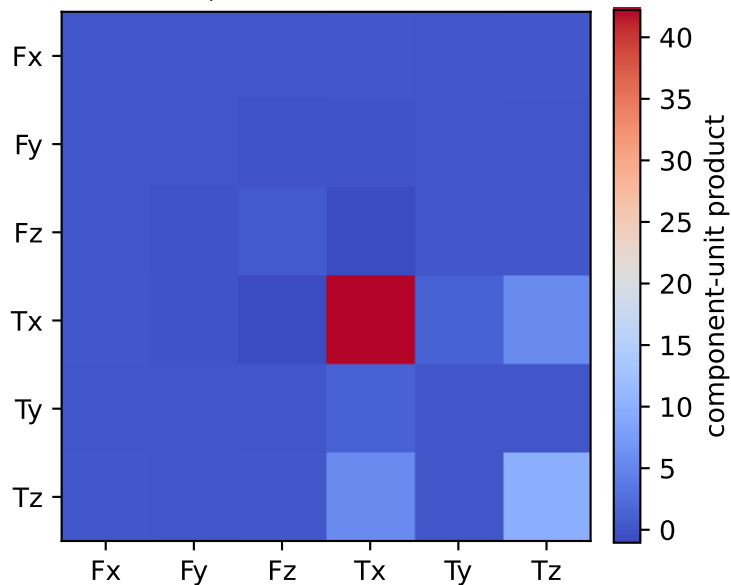
force_over_mass_rotor_2_nominal: raw Newton--Euler residual wrench (not trajectory-fitted external wrench)
mass gauge fixed to nominal mass = 2.35159759 kg



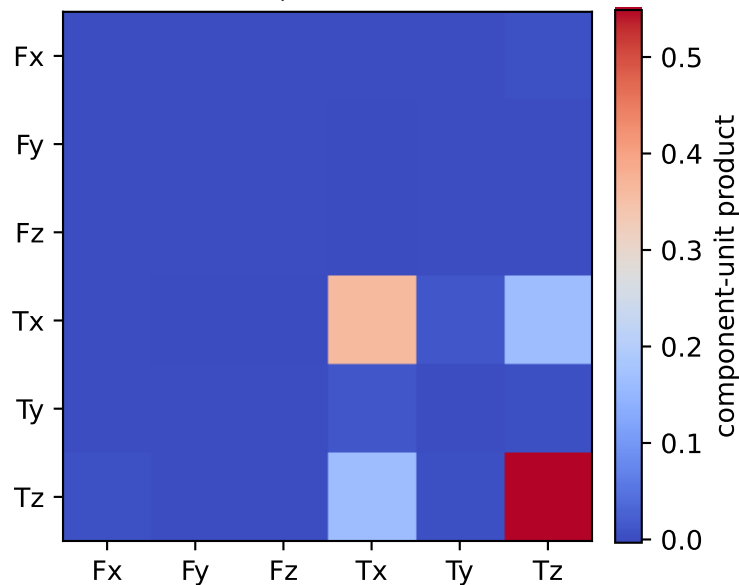
vector RMS about zero: force 0.722 N, torque 7.1972 Nm; component std about mean: force [0.2252 0.2428 0.5949], torque [6.4968 0.1976 3.1039]

force_over_mass_rotor_2_nominal: residual-wrench covariance decomposition in nominal-mass gauge
force [N], torque [Nm]; raw excess eigenvalues [6.419e-04 1.280e-02 7.440e-02 3.315e-01 8.295e+00 4.272e+01]

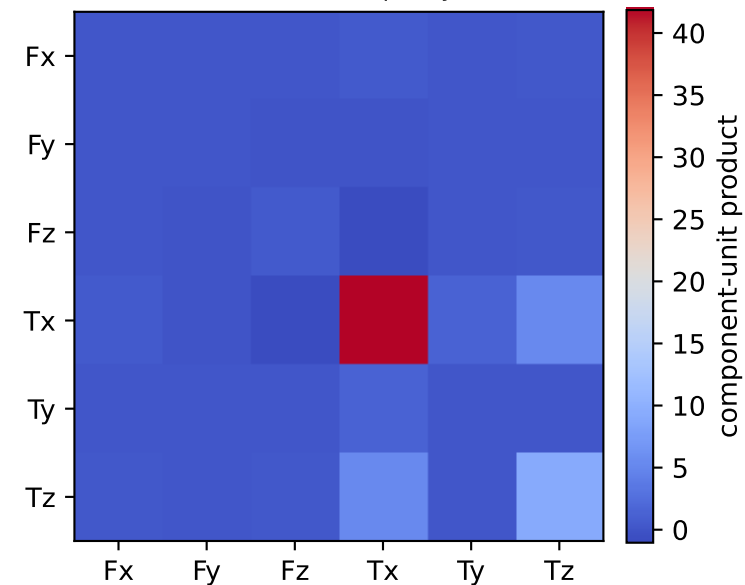
empirical total covariance



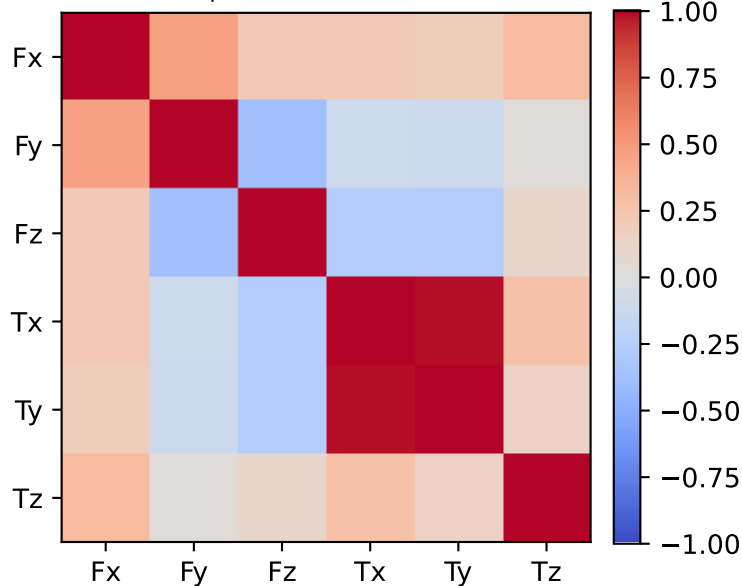
mean full-SG prediction covariance



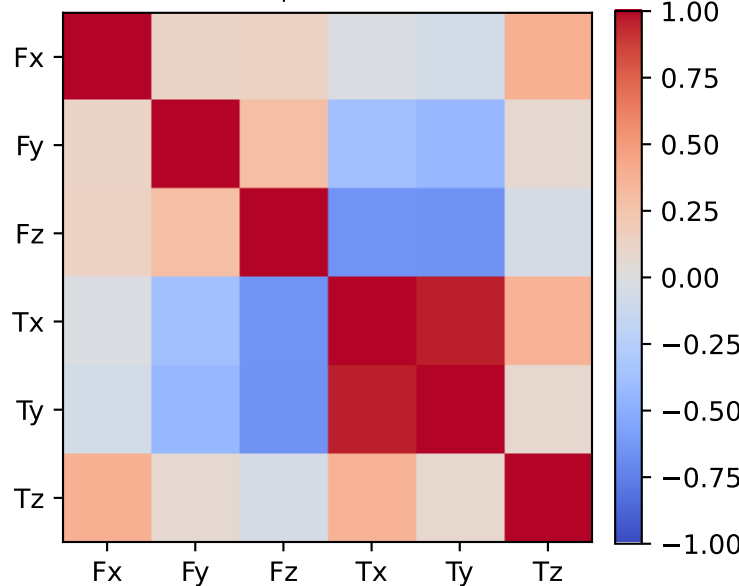
PSD excess model discrepancy covariance



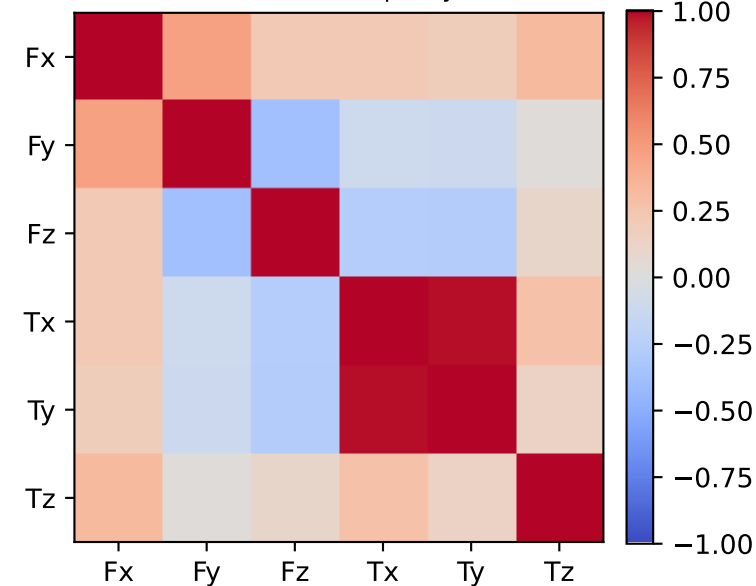
empirical total correlation



mean full-SG prediction correlation

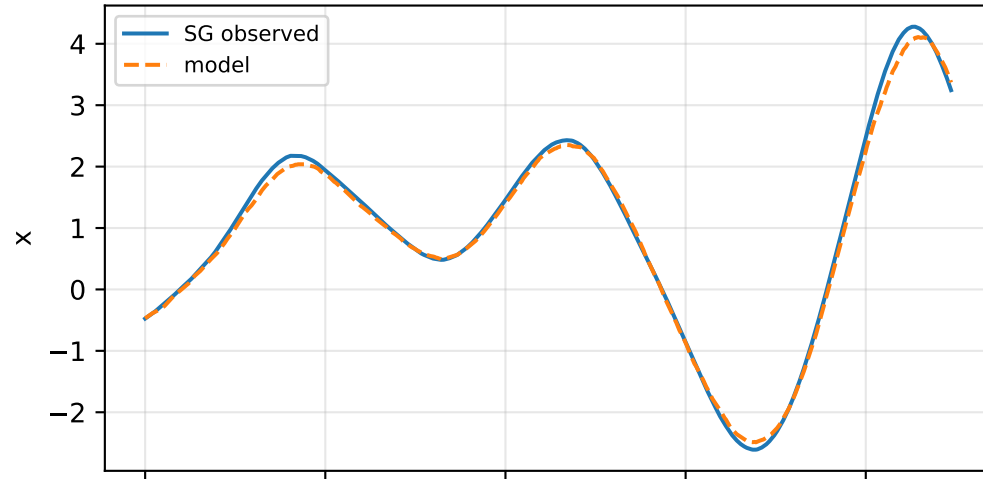


PSD excess model discrepancy correlation

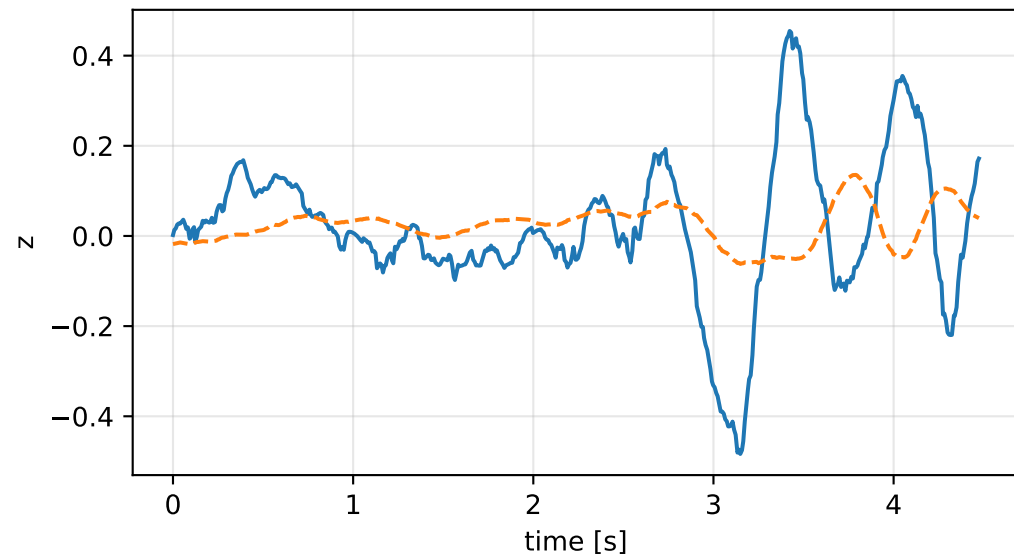
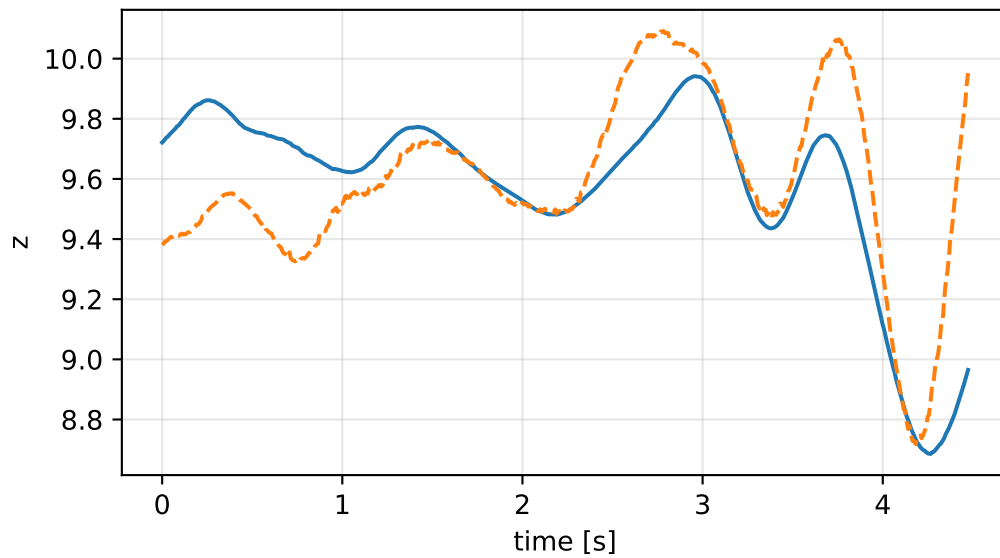
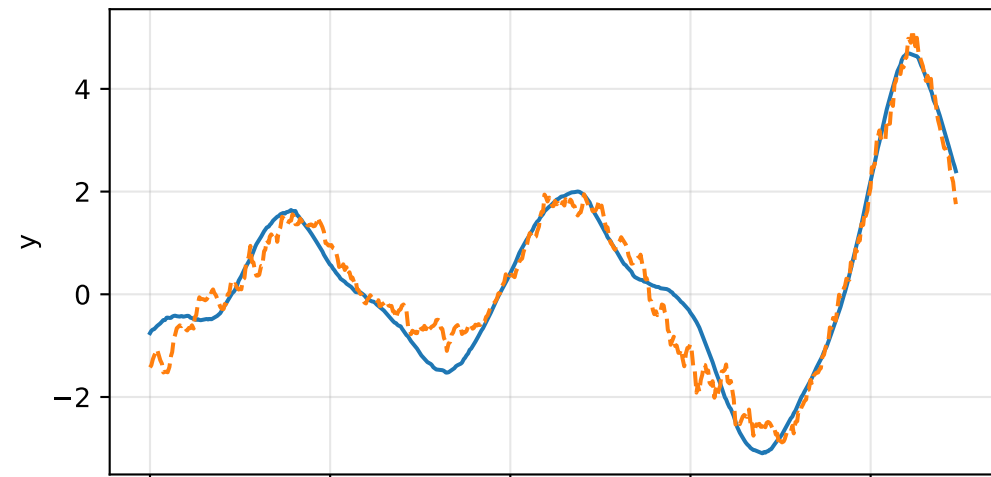
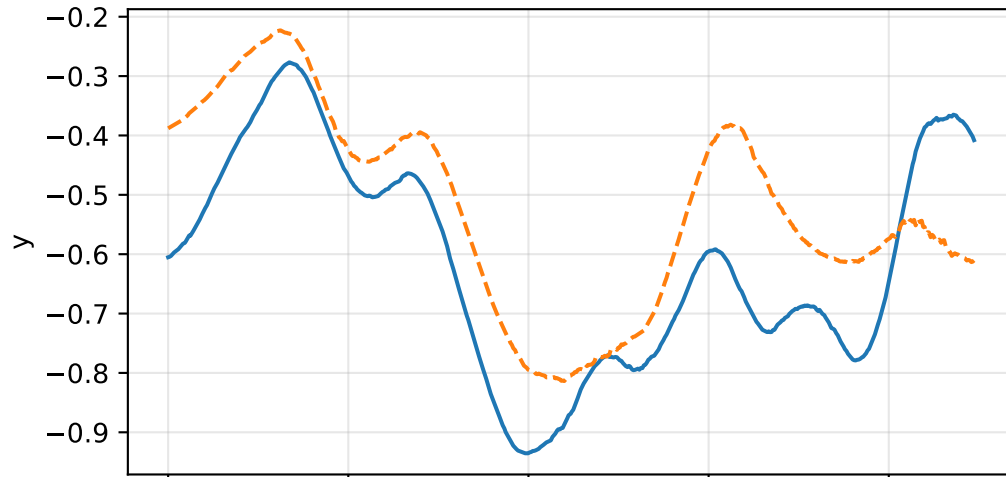
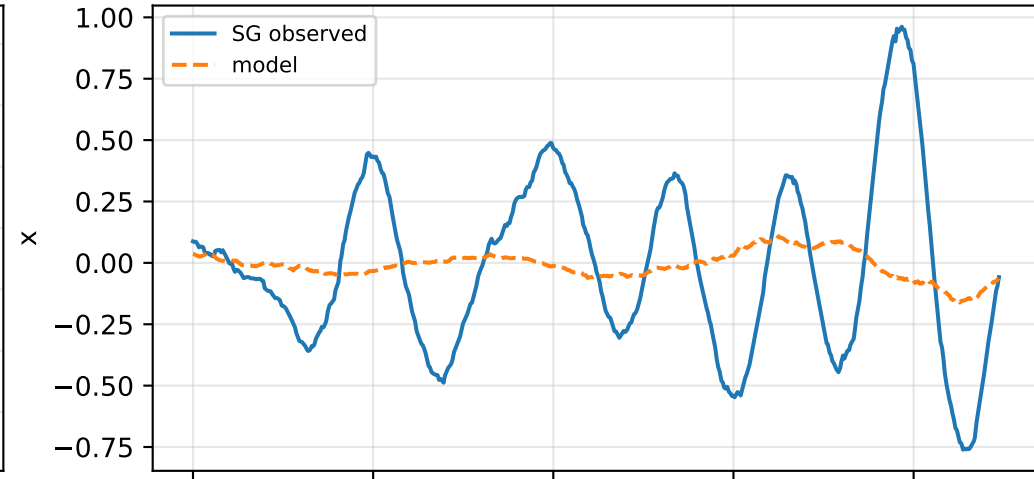


force_over_mass_rotor_2_nominal: acceleration residual objective

specific acceleration [m/s^2]

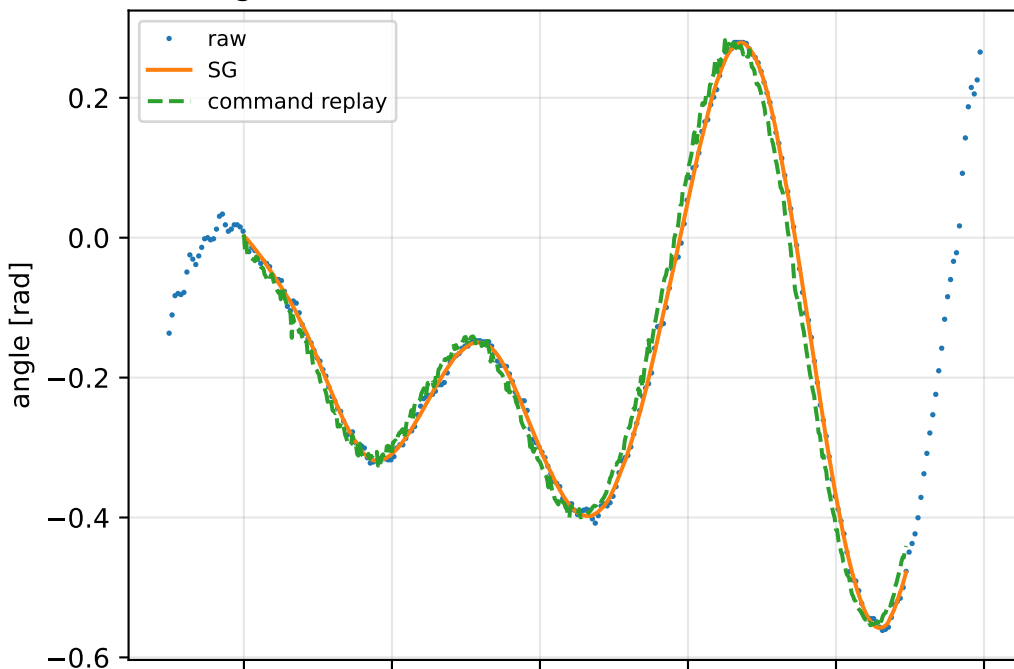


angular acceleration [rad/s^2]

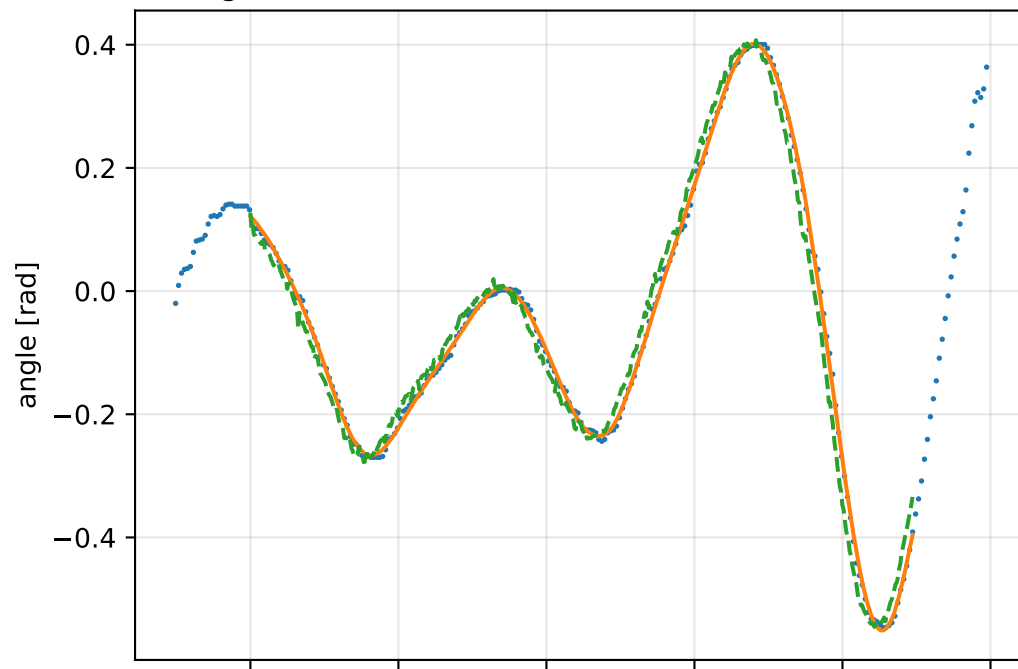


force_over_mass_rotor_2_nominal: gimbal measurement audit (objective=measured_sg)

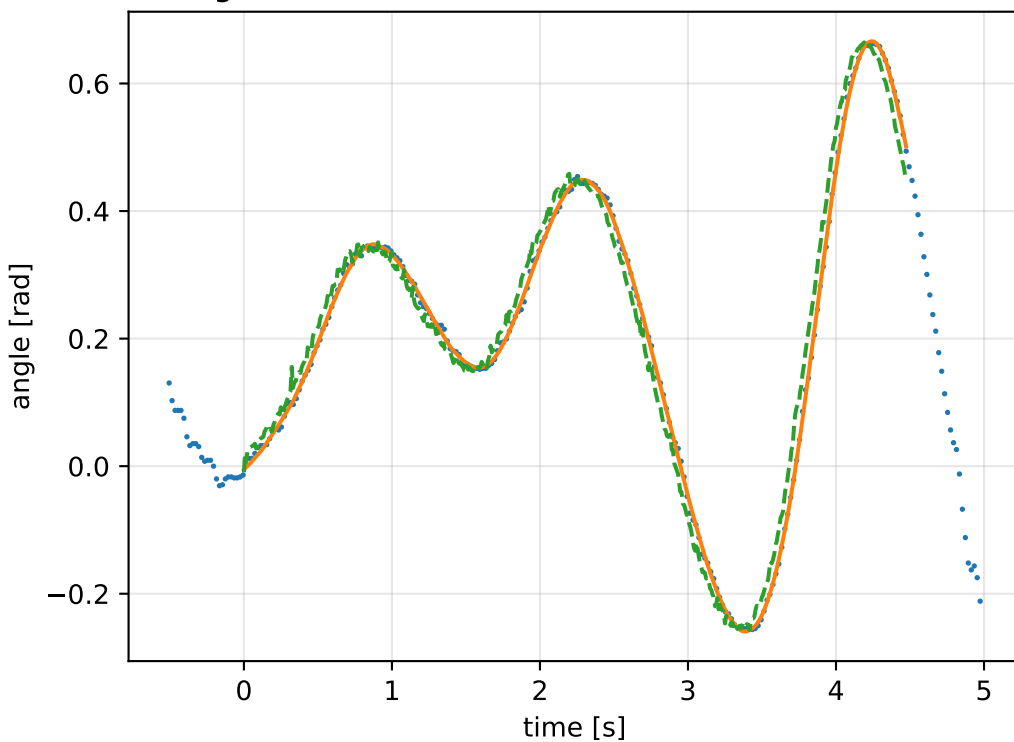
gimbal 1: RMSE=0.027, max=0.07529 rad



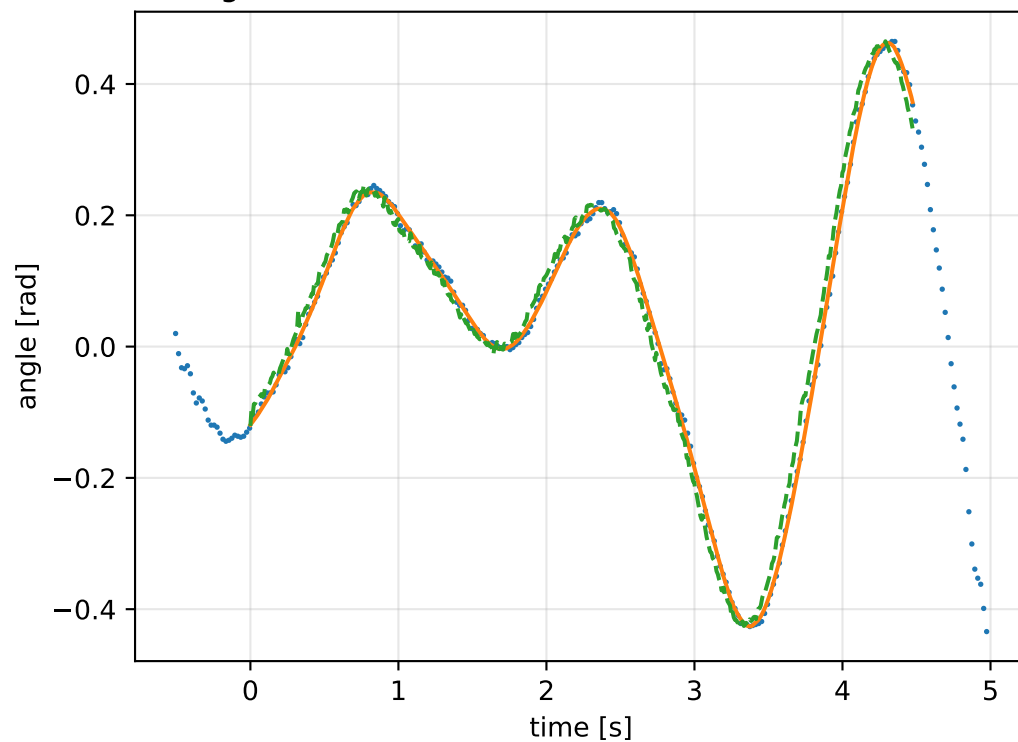
gimbal 2: RMSE=0.03107, max=0.08239 rad



gimbal 3: RMSE=0.03106, max=0.08682 rad

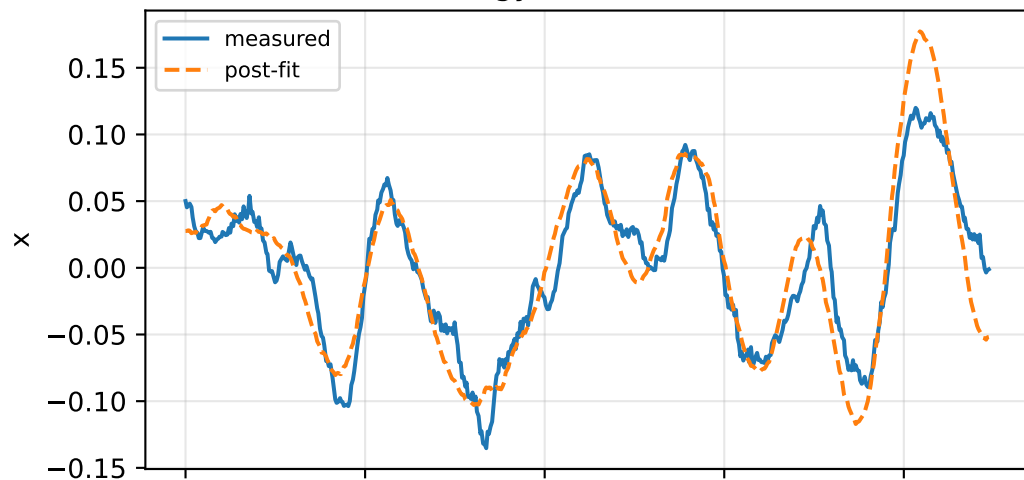


gimbal 4: RMSE=0.02598, max=0.07261 rad

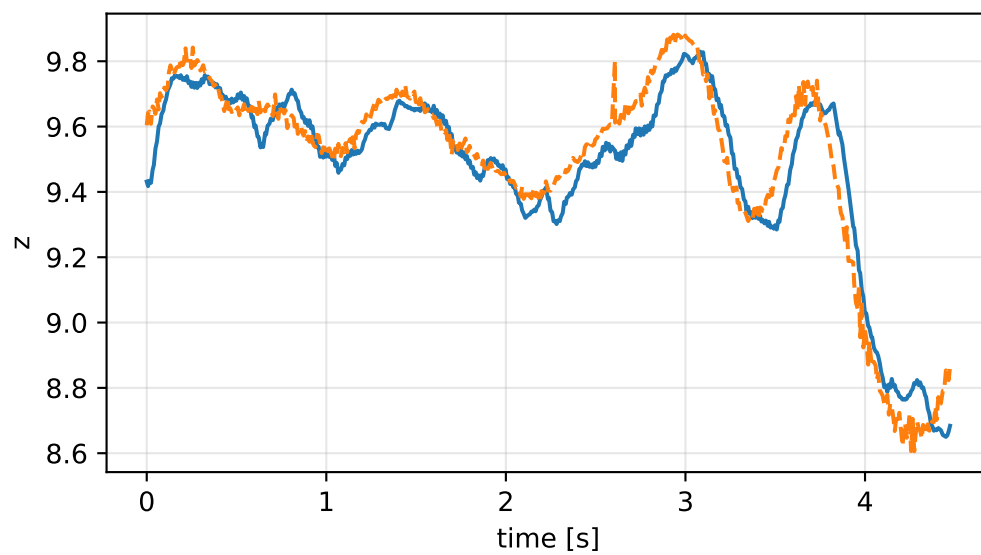
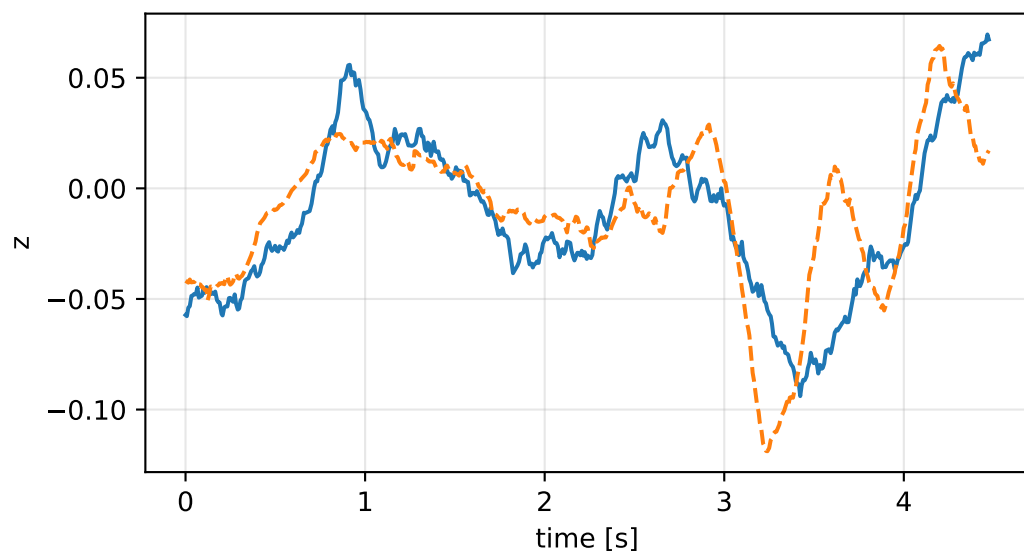
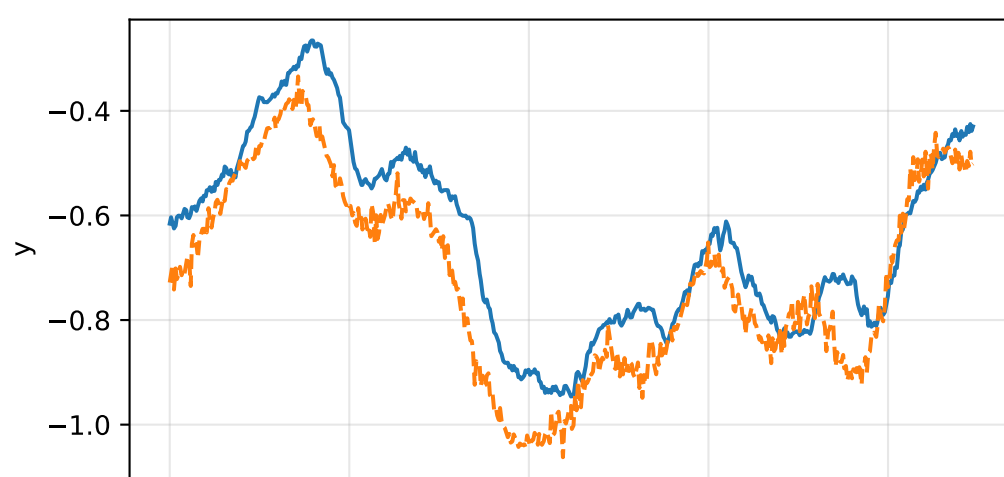
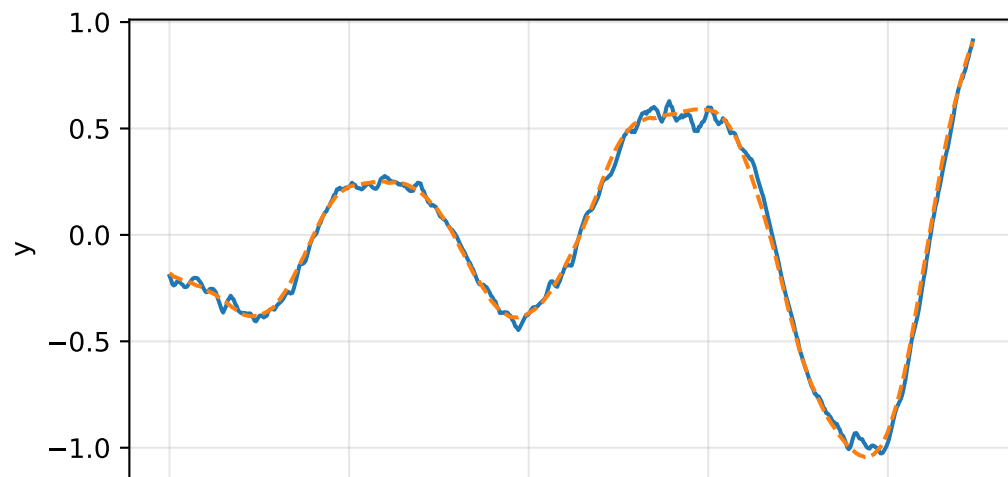
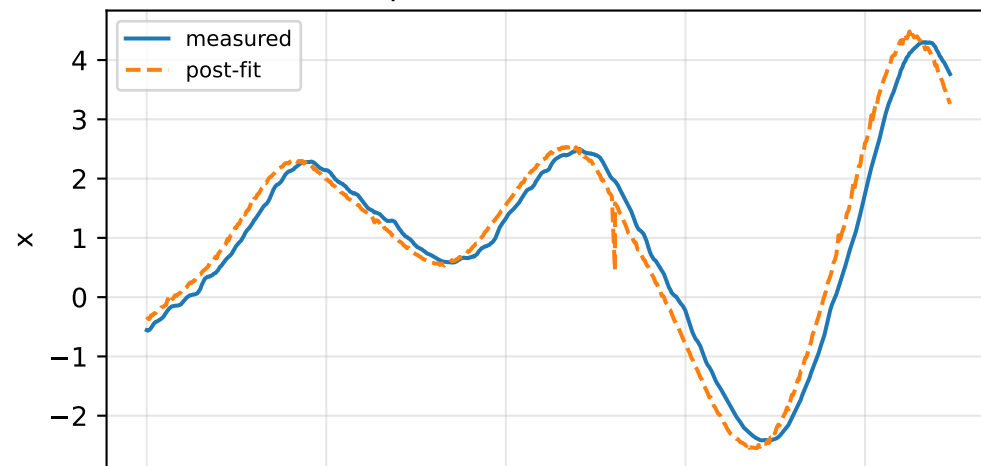


force_over_mass_rotor_2_nominal: IMU comparison (diagnostic only)

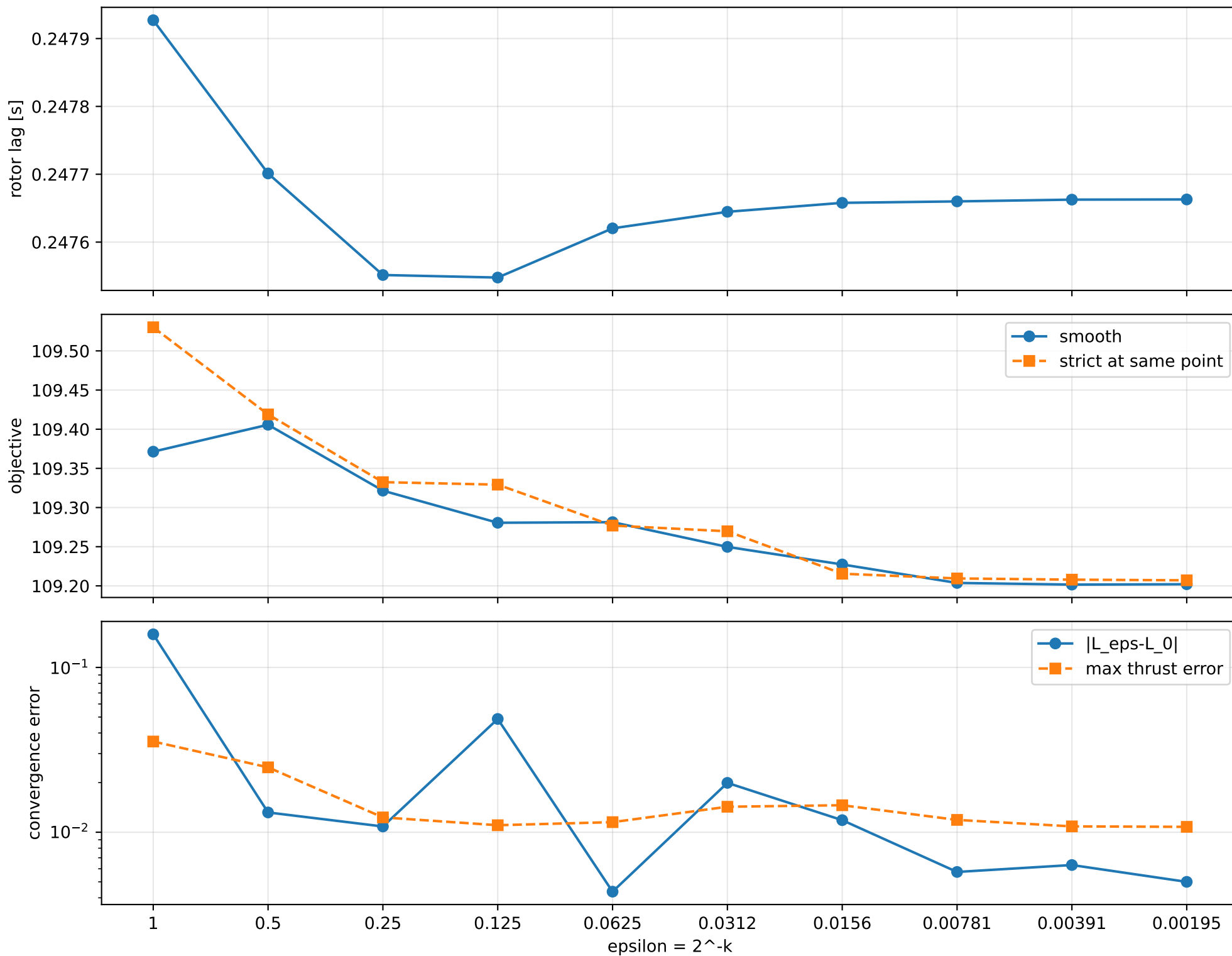
gyro [rad/s]



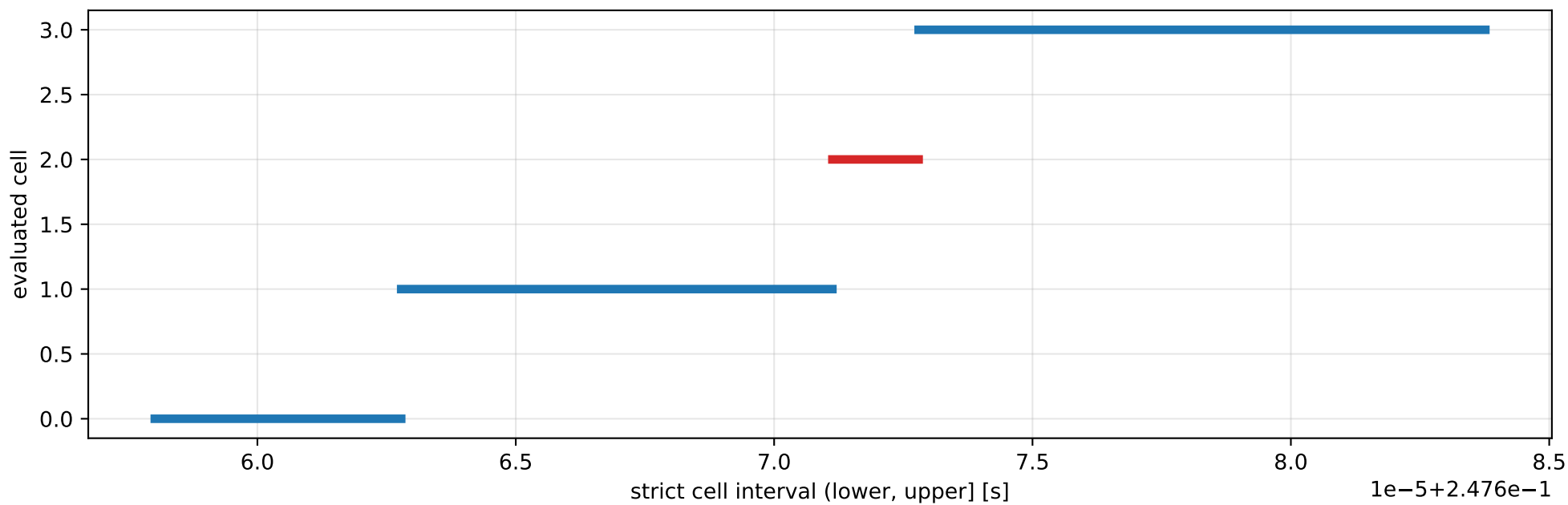
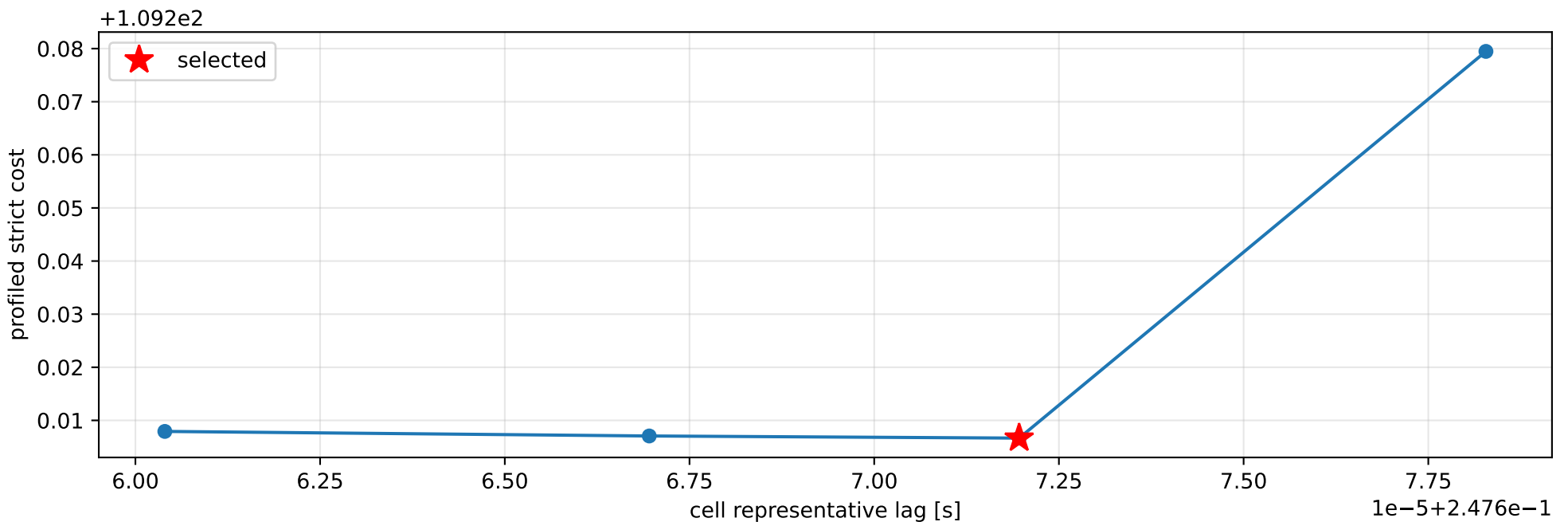
specific force [m/s^2]



force_over_mass_rotor_2_nominal: smooth-to-strict continuation



force_over_mass_rotor_2_nominal: exact strict-ZOH lag cells



selected (0.247671127, 0.247672796] s; final smooth 0.247662885 s; data boundary=False

force_over_mass_rotor_2_nominal: parameter estimate

Case: force_over_mass_rotor_2_nominal
status: completed (all stages completed)

mass [kg]: nominal 2.35159759; estimated 1.5582698

inertia [kg m²]:

```
[[12.11712008  0.38657283  0.02078827]
 [ 0.38657283  0.05767389 -0.09636654]
 [ 0.02078827 -0.09636654 12.10207664]]
```

CoG body [m]: [-0.05896075 -0.0976264 0.01015488]

force effectiveness: [0.95727014 0.66264305 0.18309985 0.43786076]

scale-free J/m [m²]:

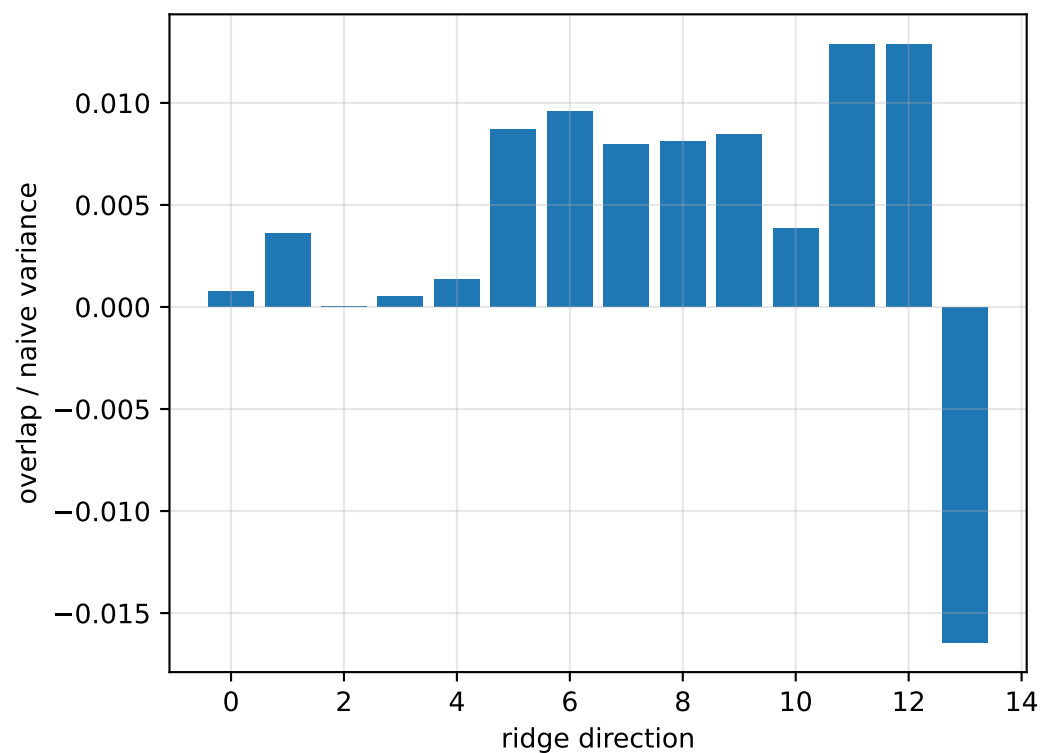
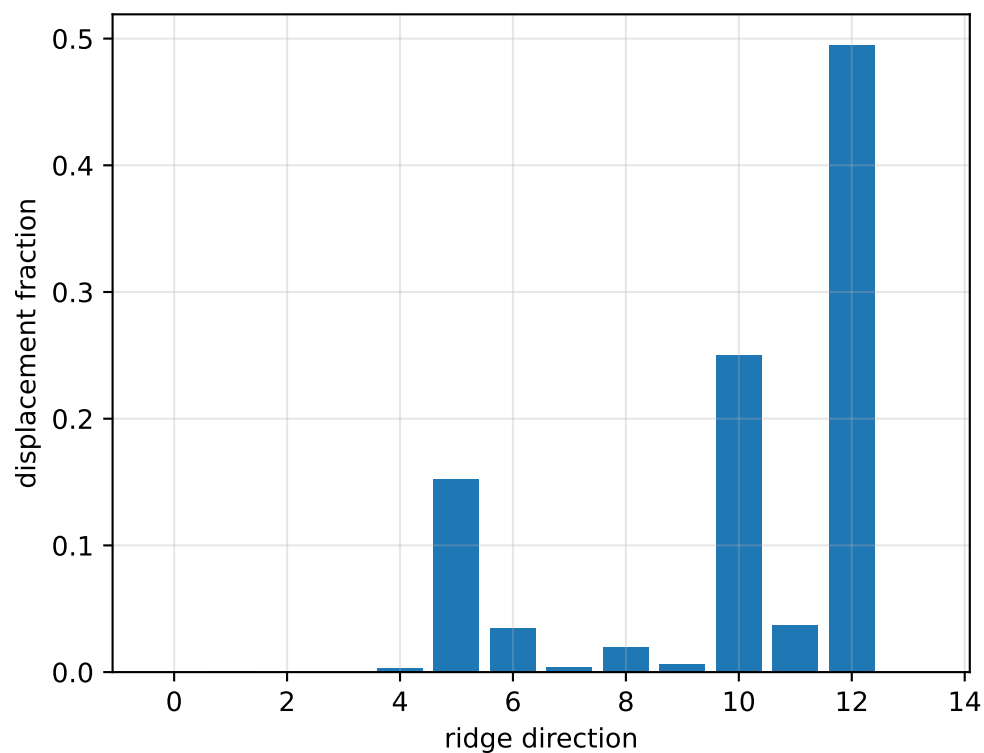
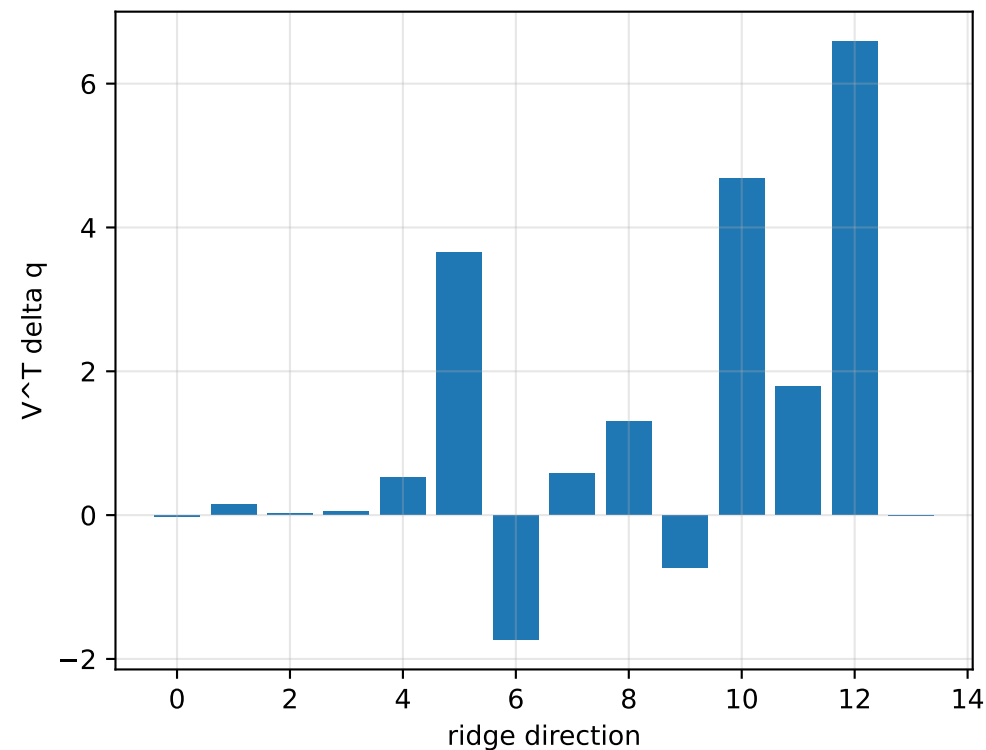
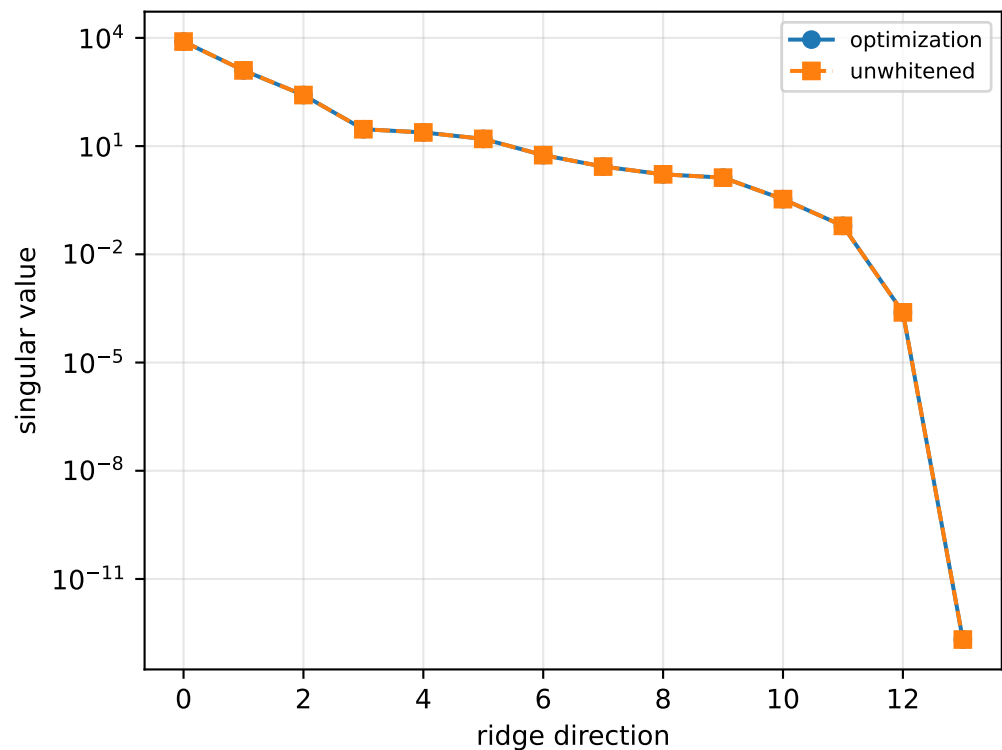
```
[[ 7.77600905  0.24807824  0.01334061]
 [ 0.24807824  0.03701149 -0.06184201]
 [ 0.01334061 -0.06184201  7.76635511]]
```

scale-free f/m: [0.61431604 0.42524283 0.11750202 0.28099162]

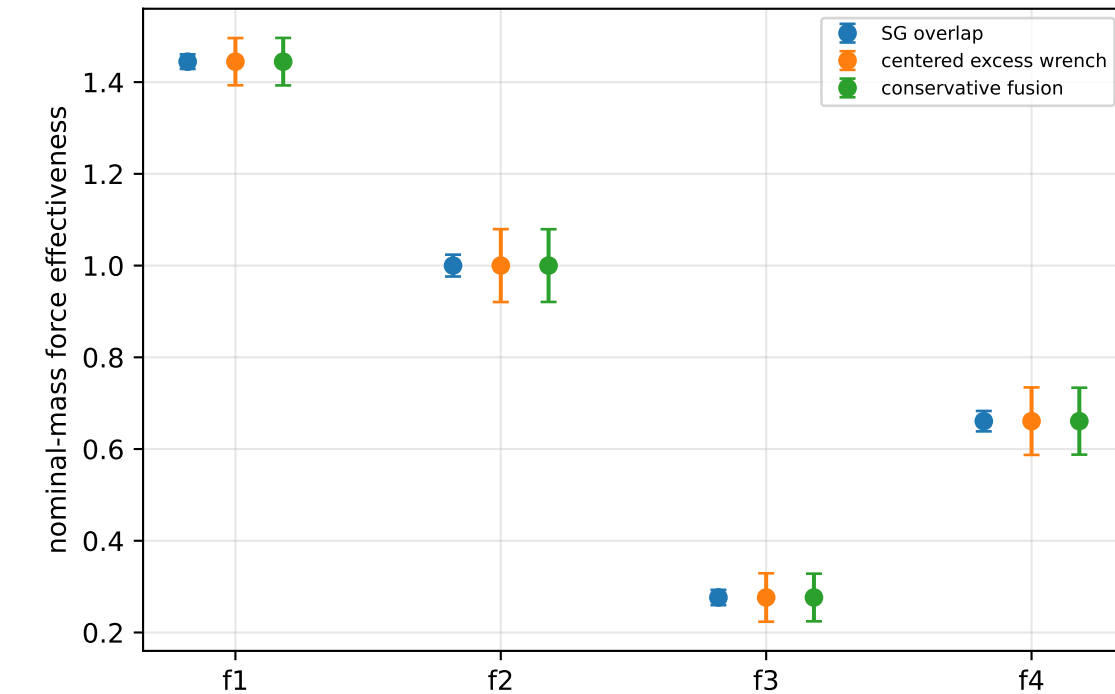
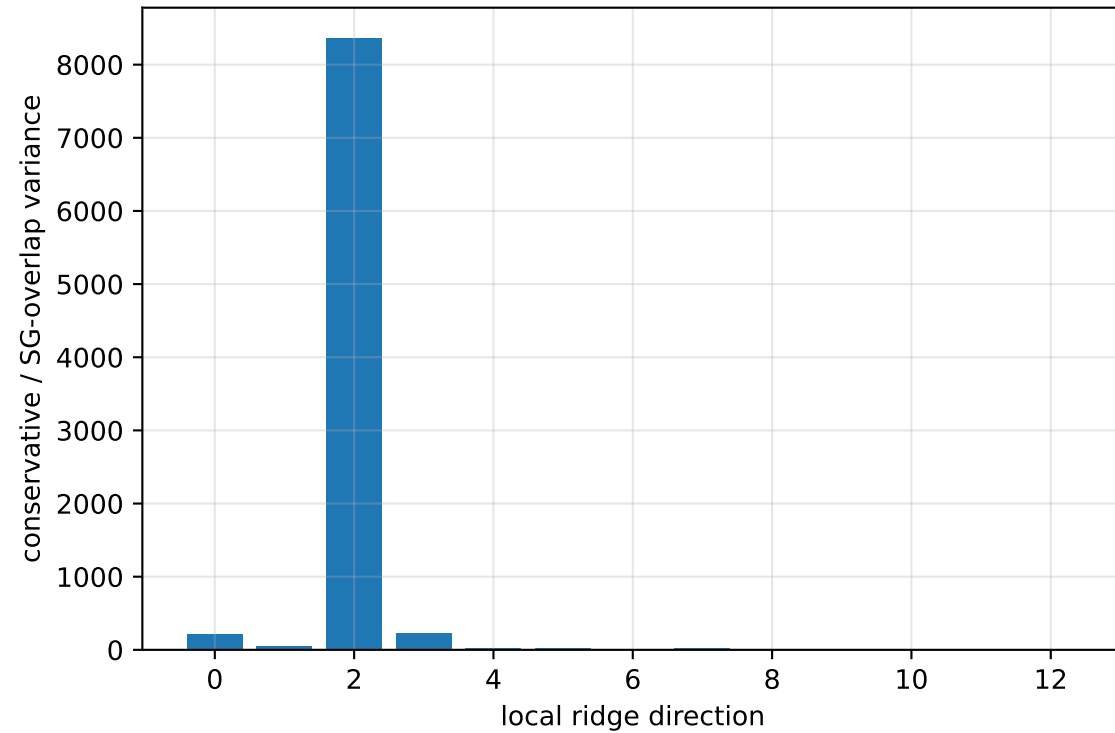
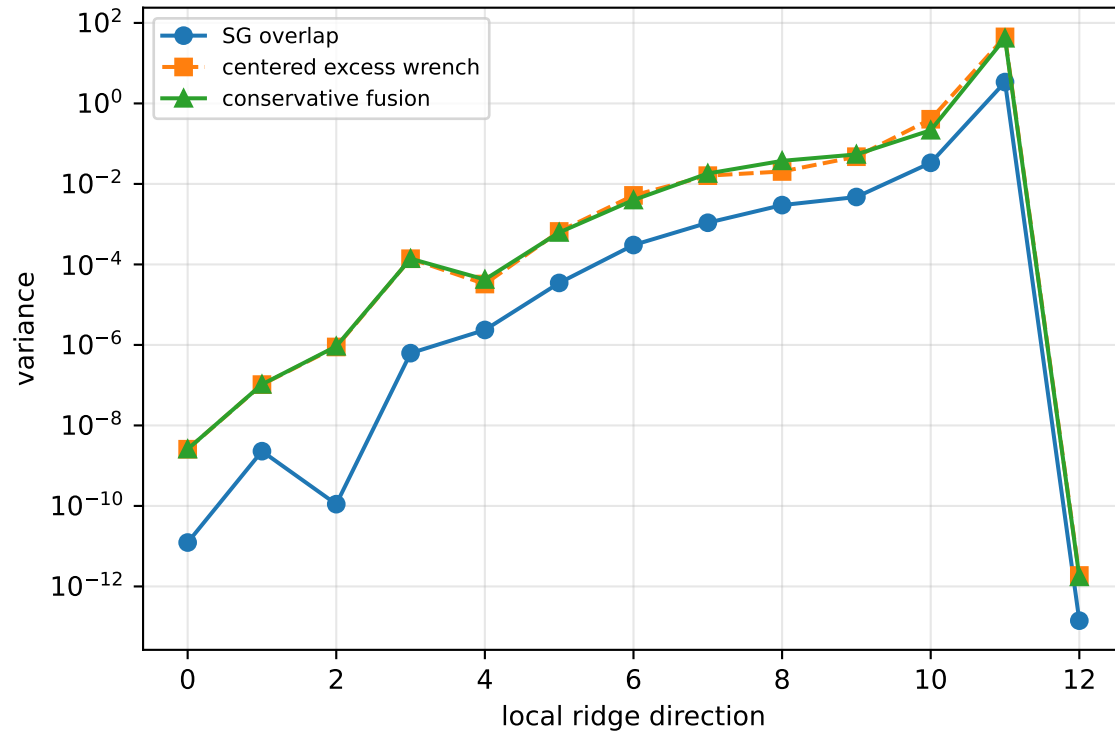
rotor lag cell: (0.247671127, 0.247672796] s

representative: 0.247671962 s

force_over_mass_rotor_2_nominal: ridge spectrum (rank 13, nullity 1, ||Jv||=3.38e-13)



force_over_mass_rotor_2_nominal: Conservative fusion uncertainty



Top three non-gauge ambiguous directions
 $\text{tr}(\mathbf{M}_{\text{res}}) / \text{tr}(\mathbf{M}_{\text{SG}}) = 191.2$
wrench-to-acceleration recovery max error = $7.11\text{e-}15$
exact scale gauge ridge index = 13 (alignment 1)

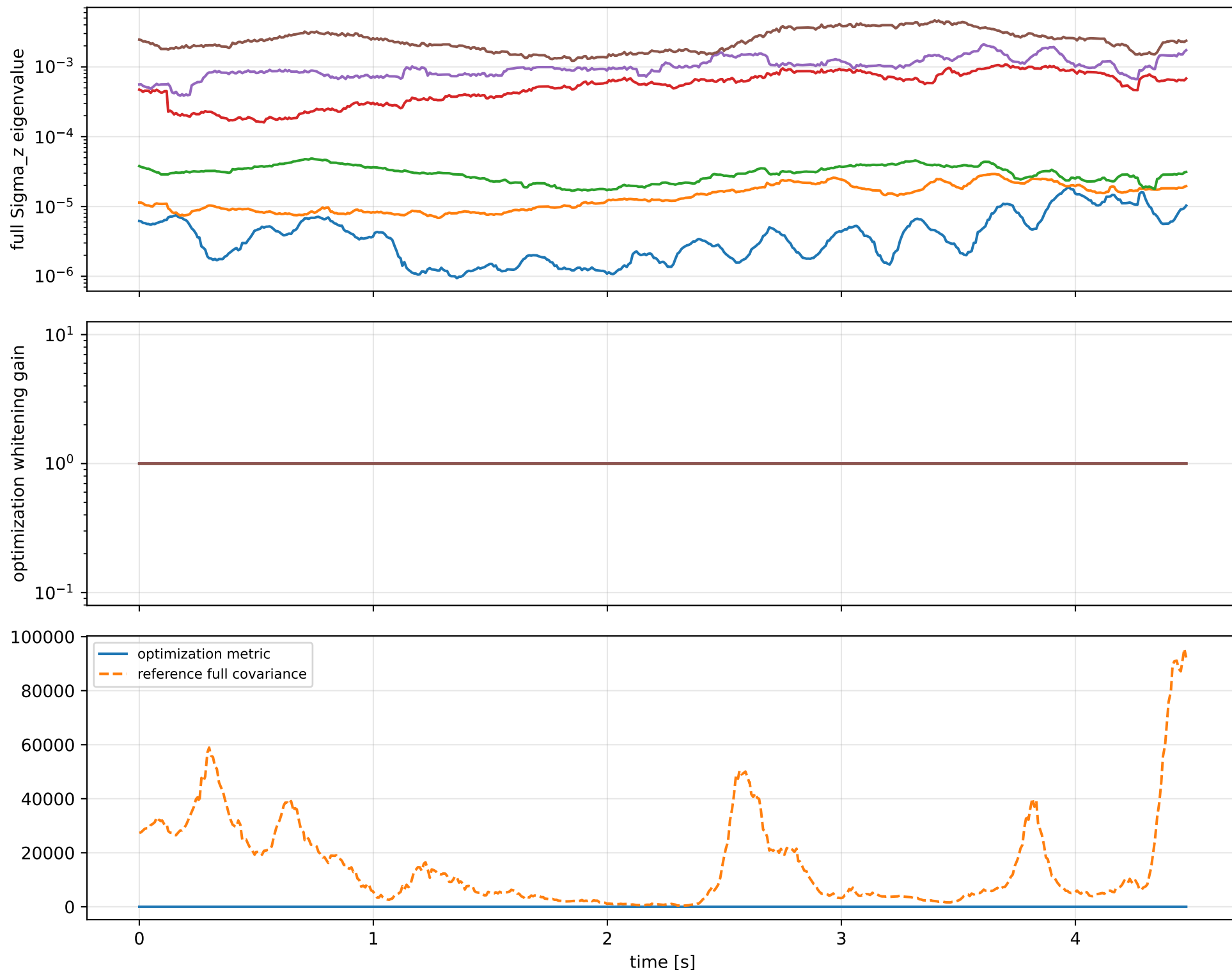
index 11: variance=41.56, ratio=12.24
 $\log_{\text{mass}} = -0.0754$; $\text{second_moment_diag_1} = -0.0473$
 $\text{second_moment_diag_2} = -0.0823$; $\text{second_moment_diag_3} = +0.507$
 $\text{second_moment_offdiag_12} = -0.073$; $\text{second_moment_offdiag_13} = +0.836$
 $\text{second_moment_offdiag_23} = -0.0281$; $\text{cog_x} = -0.00043$
 $\text{cog_y} = +0.000329$; $\text{cog_z} = +1.6\text{e-}05$
 $\log_{\text{force_eff_1}} = -0.0751$; $\log_{\text{force_eff_2}} = -0.0783$
 $\log_{\text{force_eff_3}} = -0.0774$; $\log_{\text{force_eff_4}} = -0.0713$

index 10: variance=0.2156, ratio=6.443
 $\log_{\text{mass}} = -0.153$; $\text{second_moment_diag_1} = -0.00774$
 $\text{second_moment_diag_2} = +0.847$; $\text{second_moment_diag_3} = -0.0729$
 $\text{second_moment_offdiag_12} = +0.0432$; $\text{second_moment_offdiag_13} = +0.0747$
 $\text{second_moment_offdiag_23} = +0.39$; $\text{cog_x} = -0.000631$
 $\text{cog_y} = +0.00267$; $\text{cog_z} = +0.00119$
 $\log_{\text{force_eff_1}} = -0.152$; $\log_{\text{force_eff_2}} = -0.156$
 $\log_{\text{force_eff_3}} = -0.157$; $\log_{\text{force_eff_4}} = -0.149$

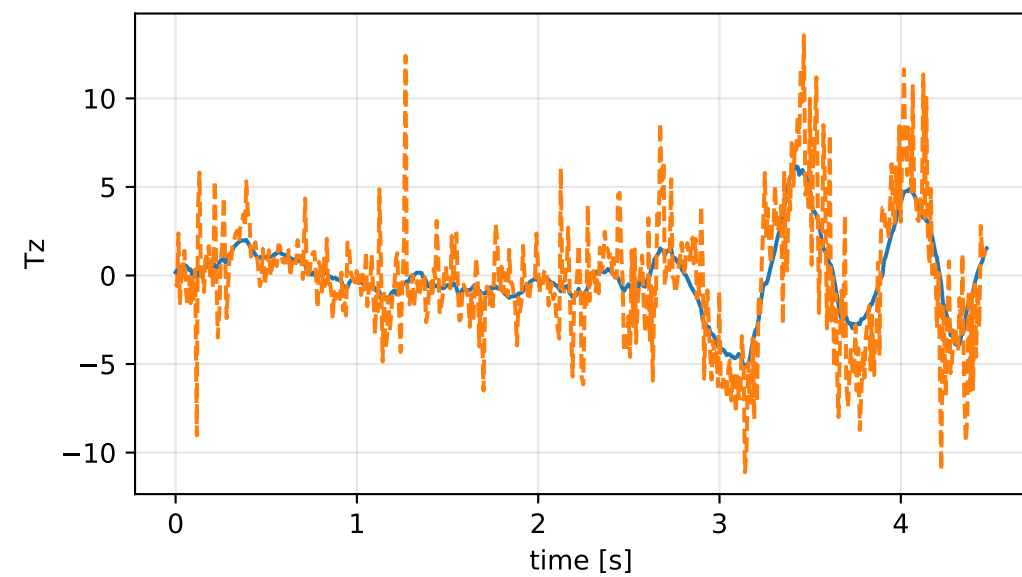
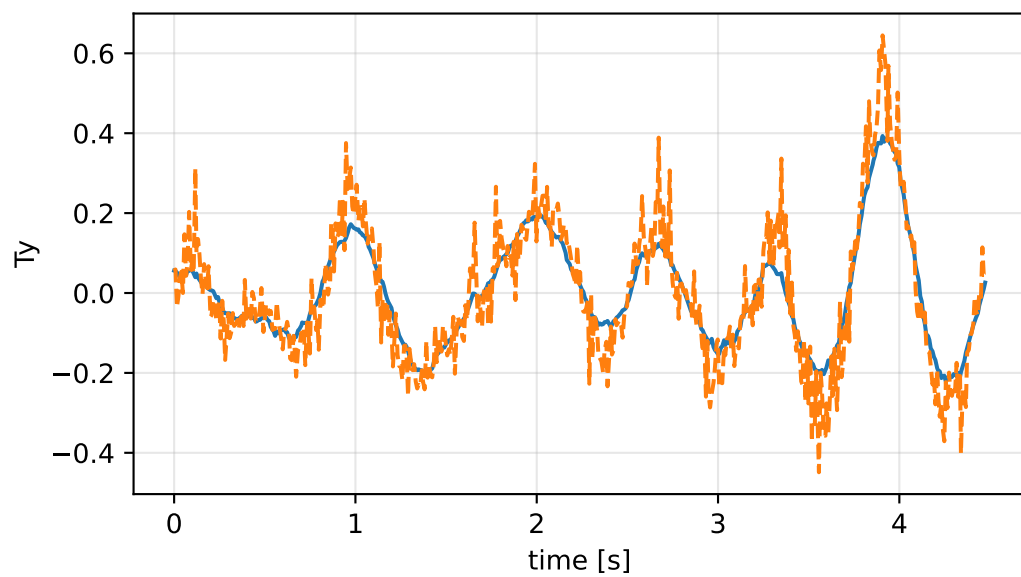
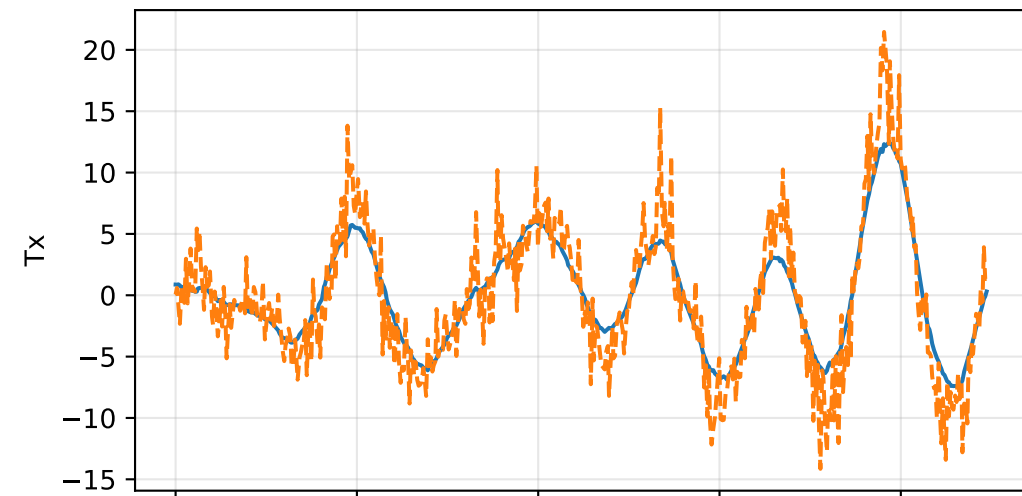
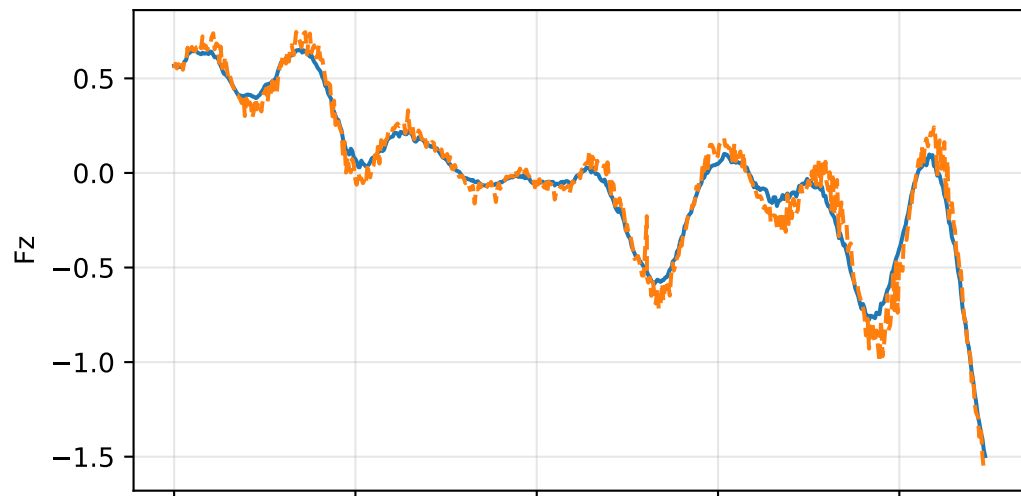
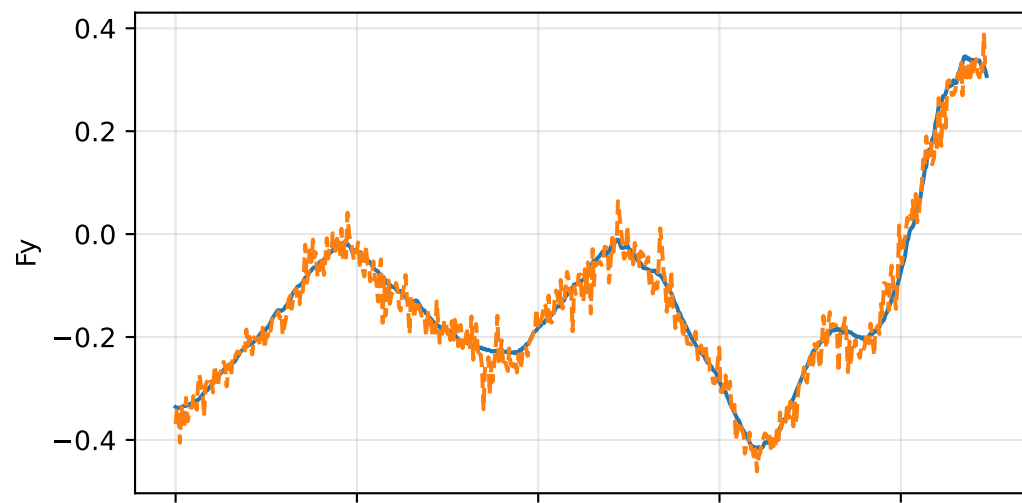
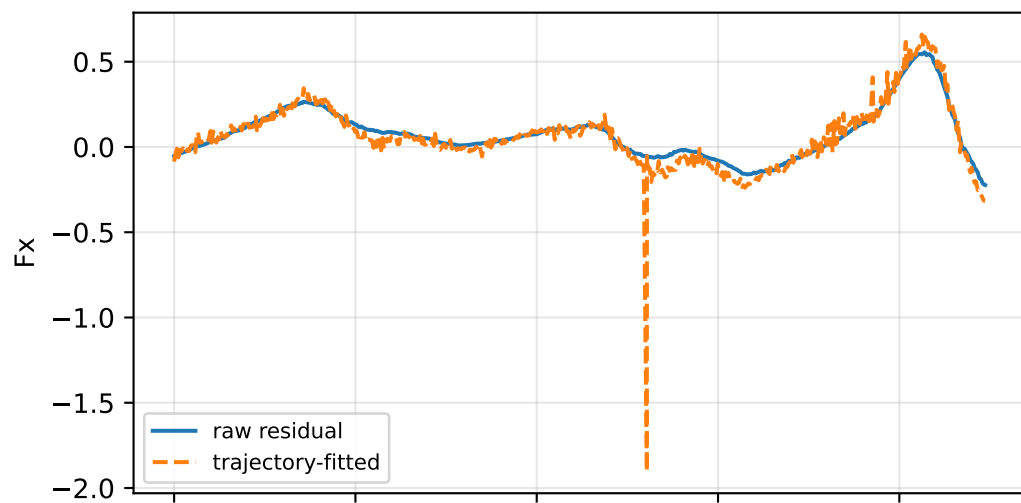
index 9: variance=0.05409, ratio=11.42
 $\log_{\text{mass}} = -0.109$; $\text{second_moment_diag_1} = -0.00408$
 $\text{second_moment_diag_2} = -0.0624$; $\text{second_moment_diag_3} = +0.08$
 $\text{second_moment_offdiag_12} = +0.149$; $\text{second_moment_offdiag_13} = -0.0327$
 $\text{second_moment_offdiag_23} = +0.151$; $\text{cog_x} = +0.071$
 $\text{cog_y} = -0.0121$; $\text{cog_z} = +0.00579$
 $\log_{\text{force_eff_1}} = -0.261$; $\log_{\text{force_eff_2}} = +0.222$
 $\log_{\text{force_eff_3}} = +0.7$; $\log_{\text{force_eff_4}} = -0.565$

The conservative fusion covariance deliberately retains the existing SG-overlap uncertainty and adds the uncentered empirical residual score second moment without subtracting the SG contribution. It is intended as a conservative fusion distribution, not as a calibrated generative noise covariance.

force_over_mass_rotor_2_nominal: optimization vs reference covariance



force_over_mass_rotor_2_nominal: wrench / closure (force max 4.77e-15, torque max 3.55e-15)



force_over_mass_rotor_2_nominal: optional parameter-prior audit

Optional physical parameter prior

The parameter prior is optional external information. The default estimator is prior-free.

name: pseudo_condition_force_over_mass_rotor_2_nominal

role: pseudo_conditioning_ablation

source: /home/leus/catkin_ws/src/ProbTF-demo/ros/examples/grape-param-estim/minimal/config/priors/pseudo_conditioning/scalar/force_over_mass

SHA256: 4ad6b24989a0df886676644a499e15946486f27db58e2dfd259bd888c891980e

data objective: 109.206662386

prior objective: 1.66369356062e-08

total objective: 109.206662403

This configuration uses an intentionally tight artificial standard deviation for an ablation-style pseudo-conditioning experiment; it is not a calibrated physical uncertainty.

factor: force_over_mass_rotor_2_nominal (force_effectiveness_over_mass)

components: rotor_2

target: [0.42524282]

std: [1.e-05]

final: [0.42524283]

error: [1.82411269e-09]

standardized residual: [0.00018241]

factor objective: 1.66369356062e-08